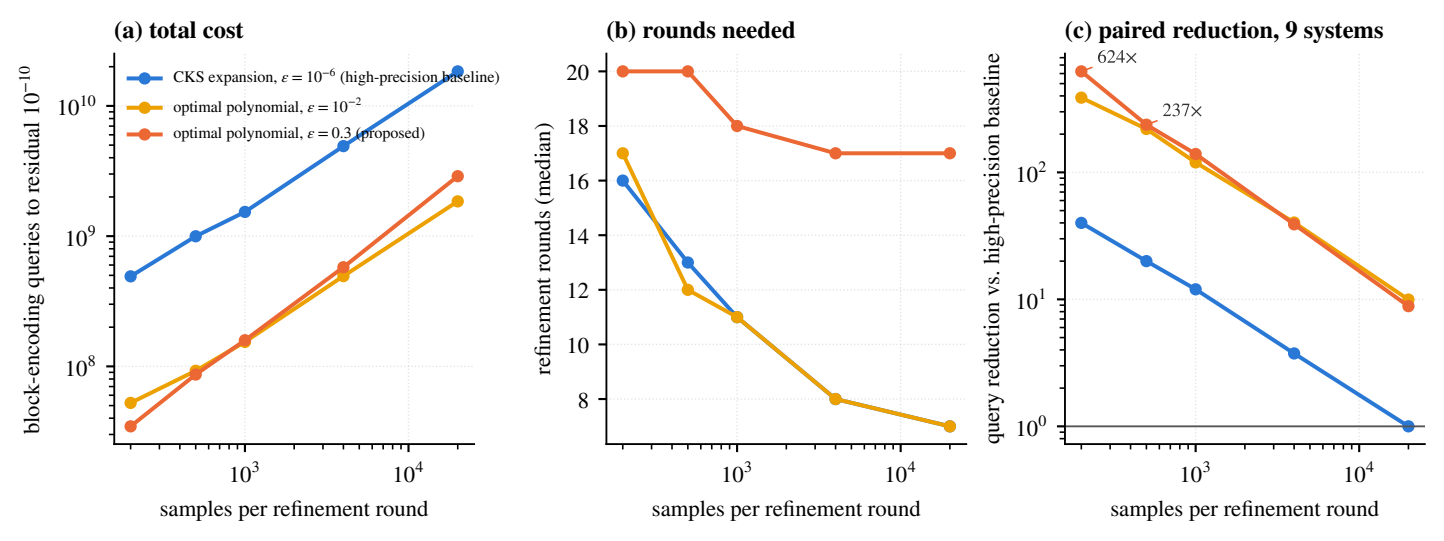
(a) total cost
CKS expansion, $\varepsilon = 10^{-6}$ (high-precision baseline)
optimal polynomial, $\varepsilon = 10^{-2}$
optimal polynomial, $\varepsilon = 0.3$ (proposed)
block-encoding queries to residual 10^{-10}
10^{10}
10^{9}
10^{8}
10^{3}
10^{4}
samples per refinement round
(b) rounds needed
refinement rounds (median)
20
18
16
14
12
10
8
10^{3}
10^{4}
samples per refinement round
(c) paired reduction, 9 systems
query reduction vs. high-precision baseline
624×
237×
10^{2}
10^{1}
10^{0}
10^{3}
10^{4}
samples per refinement round